

Q1) Find the datatype of these two declarations.

```
x = 5
y = "John"
```

```
x=5
y="John"
print(type(x))
print(type(y))
```

```
<class 'int'>
<class 'str'>
```

Q2) Check whether the following syntax is **valid** or **invalid** for naming a variable. ?

Example: `abc=100` #valid syntax

- 1) `3a=10`
- 2) `@abc=10`
- 3) `a100=100`
- 4) `_a984_=100`
- 5) `a9967$=100`
- 6) `xyz-2=100`

```
3a=10
print(3a)
```

```
Cell In[25], line 1
    3a=10
    ^
SyntaxError: invalid decimal literal
```

```
@abc=10
print(@abc)
```

```
Cell In[29], line 1
    @abc=10
    ^
SyntaxError: invalid syntax.
```

```
_a984 =100  
print(_a984_)
```

100

```
a100=100  
print(a100)
```

100

```
a9964$=100  
print(a9964$)
```

```
Cell In[35], line 1  
a9964$=100  
      ^
```

SyntaxError: invalid syntax

```
xyz-2=100  
print(xyz-2)
```

```
Cell In[37], line 1  
xyz-2=100  
      ^
```

SyntaxError: cannot assign to expression here. Maybe you meant '==' instead of '='?

Q3) Check if an element exists in the list in Python.

list = test_list = [1, 6, 3, 5, 3, 4]

1) Check if **3** exist or not.

2) Check if **9** exists or not.

```
# Define the list  
test_list = [1, 6, 3, 5, 3, 4]  
  
# Check if 3 exists in the list  
if 3 in test_list:  
    print("3 exists in the list.")  
else:  
    print("3 does not exist in the list.")  
  
# Check if 9 exists in the list  
if 9 in test_list:  
    print("9 exists in the list.")  
else:  
    print("9 does not exist in the list.")
```

3 exists in the list.

9 does not exist in the list.

Q4) Take the user **input** to print the current date?

```
import datetime

# Take user input (press Enter to proceed)
input("Press Enter to print the current date...")

# Get the current date
current_date = datetime.date.today()

# Print the current date
print("The current date is:", current_date)
```

Press Enter to print the current date...
The current date is: 2024-09-15

Q5) What is the **output** of the following code?

- a) print 9//2
- b) print 9%2

```
print (9//2)
print (9%2)
```

4
1

Q6) Print the **First 10 Natural Numbers** using a while loop.

```
# Initialize the variable
i = 1

# While loop to print first 10 natural numbers
while i <= 10:
    print(i)
    i += 1
```

1
2
3
4
5
6
7
8
9
10

Q7) Write a program to accept a **number from a user** and calculate the **sum of all numbers from 1 to a given number**.

Example= if the user entered **10** the output should be **55 (1+2+3+4+5+6+7+8+9+10)**

```
# Accept a number from the user
n = int(input("Enter a number: "))

# Initialize the sum
total_sum = 0

# Calculate the sum of all numbers from 1 to n
for i in range(1, n + 1):
    total_sum += i

# Print the result
print(f"The sum of all numbers from 1 to {n} is: {total_sum}")
```

Enter a number: 10

The sum of all numbers from 1 to 10 is: 55

Q8) Write a Python program that iterates the integers from **1** to **50**. For multiples of **three** print "**Fizz**" instead of the number and for multiples of **five** print "**Buzz**". For numbers that are multiples of both **three** and **five** print "**FizzBuzz**"?

Example:

FizzBuzz

1

2

Fizz

4

Buzz

```
# Iterate through numbers from 1 to 50
```

```
for i in range(1, 51):  
    if i % 3 == 0 and i % 5 == 0:  
        print("FizzBuzz")  
    elif i % 3 == 0:  
        print("Fizz")  
    elif i % 5 == 0:  
        print("Buzz")  
    else:  
        print(i)
```

```
1  
2  
Fizz  
4  
Buzz  
Fizz  
7  
8  
Fizz  
Buzz  
11  
Fizz  
13  
14  
FizzBuzz
```